Web Development Lab 3

CS 1301 - Intro to Computing - Spring 2026

Important

- Due Date: Reference the assignment on Canvas.
- You will collaborate with a partner for this and the following lab
- **Lab 4 builds directly on Lab 3 so that both labs produce one final web application**
- You will be using the same API for Lab 3 and Lab 4, so choose wisely!
- You may deploy your app at any point during this lab. However, by the end of the lab, your local files should match the version of the app that is deployed.
- There will be a **Peer Review** at the end of the lab. Poor peer review feedback may result in a letter grade deduction.
- Only **one submission per team** is required on Gradescope.
- Cross-group collaboration is **NOT allowed**.
- **Please read the entire document before starting this assignment.**

Document Overview

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Overview

As a brief summary, your web application will contain the following pages:

- A home page to breifly explain the purpose of your app and support a multi-page setup

- An analytical or creative app that processes data fetched from a web API of your choice.
- An incorporation of Google Gemini to make a Chatbot

Lab Rules

You may **not** use any kind of generators or website/app builders other than streamlit itself.
All code written in this assignment must be written by you. While we encourage the use of Google to reference streamlit properties, any **large sections** of code taken from another source are not permitted (including the projects listed on Streamlit's website).

Purpose

For this lab, you will build an additional web application using the `Streamlit` Python libary, which can easily transform Python code into interactive web applications, dashboards, or reports that can be hosted online. Enjoy!

Getting Started

Starter Files

To start this lab, download the `Lab03.zip` file from Canvas. Extract the files by right clicking the `.zip` and clicking "Extract All" (for Windows) or double-click the file on Mac. Once you extract the files, you should see the following file structure:

```
Lab03
    - requirements.txt # File for all modules that have to be installed
    - Images (folder) # Container for your images
    - Home_Page.py #File for your home page to support multi page setup
        - pages #folder for your sidebar pages
```

Use these files as a base to start of your lab. Make sure that all images you add from your computer are inside the **images** folder shown above. **All files related to your lab must be in this Lab03 folder.**

Resources

- Installing Streamlit Handout
- TA Helpdesk

- Ed Discussion
- Streamlit's official documentation is the most comprehensive and authoritative source for learning Streamlit. It provides in-depth guides, tutorials, and examples, making it the perfect starting point for beginners and advanced users. Documentation
- Streamlit's official YouTube channel is a valuable resource for visual learners. It features video tutorials and demonstrations that help you understand how to create web apps with Streamlit. Channel

  Additionally, the Lab Team have implemented an **EXAMPLE** of Lab 3 to help inspire students. To ensure that your work is not too similar to our own, do not use the poke-api for data. Only the **Pokedex** and **PokeBot** pages are required for this lab. Keep in mind that the PokeBot page incorporates data from the poke-api which is not the requirement for this lab. Check it out here Link: https://cs1301webdevlab03.streamlit.app/
- **This assignment is meant to be a self-guided exploration of web development. You are encouraged to search resources independently before seeking TA help.**

Page 1: Processing Web API Data

On Page 1, build a streamlit page by analyzing, visualizing, or presenting data fetched from an external web API. You have full creative freedom to make your project content about whatever you want, as long as the content is school appropriate. Your goal for this part of the lab should be to create a **functional** app that serves the purpose of your choice. Feel free to look at the Canvas Module. Your page must meet the following requirements:

- Must have a clear and meaningful topic or purpose.
- Must incorporate the analysis or presentation of data from an **external web API** into the application. You may display relevant data from your API of choice, or provide functionality related to your API. For example, a weather API can be used to display location-specific weather data, or a finance API can be used to display real-time stock price information.
  - You must analyze and process data received from your API by yourself in code. You may **not** use an LLM to assist with any processing or analyzing of the data . You **can not** use the Pokemon API (the same API as the example).
- Must allow at least 2 user inputs or interactions that **affect the app's behavior** (e.g., text input, sliders, select boxes, etc.). Buttons, sliders, etc. with no purpose will not be counted.
- Must contain at least two dynamic visual representation (e.g: images, graphs, plots) that **changes depending on the user inputs**.
  - For example, images can be displayed that depend on what is retrieved from the API, or graphs can be displayed from API data.

- At least one **Dynamic/Interactable** Chart or graph displaying relevant information from your API
- If you use modules that are not part of the Python Standard Library (`math` , `random` , etc.), you **must** list out all of the modules that must be installed for your app to work in `requirements.txt` , one module per line. Streamlit is already included in the `requirements.txt` file that is provided.
  - Do not include the `os` module here, since `os` is a module that is a part of the Python Standard Library.

These requirements are the minimum. Feel free to add anything else. **Be creative!** You will use the same web API for further data processing and analysis in Lab 4.

Page 2: AI Chatbot Implementation

Build another page that contains a chatbot that can sustain an intelligent and functional conversation. This page should ***NOT*** pull any data from your API from Page 1.

Hint: Check the Canvas modules for help on how to implement this!

- Must include error handling (Try/Except) for errors that may be raised by the LLM. This may include hitting rate limits due to submitting too many LLM requests at once, or accidentally generating inappropriate content. Your app must never crash due to errors raised by the Google Gemini LLM API.

- Must be able to remember prior conversation
- Must keep with the same theme from your API data (again, it should not pull data from your API)

Please do not be overwhelmed if you are intimidated by these requirements. Visit the Canvas Pages for more details.

Submission Process and Demo

Submit a **zipped** file with all the requirements to Gradescope by the deadline. Failure to do so will result in a **0**. You will need to submit a link to your deployed web application as well as all files used in Lab 3 (all images and python files). To do this, compress the entire Lab 3 folder that you are working in into a `.zip` file. Follow the instructions below:

Note: Only one submission is required per group. Following the submission process, you may add your partner onto your submission, hence creating a group submission. More instructions can be found under the Canvas page titled `Lab Team Verification Explanation`.

1. ***Zip File Submission***

i. Right click your Lab 3 folder:

- On Windows, click **Send To > Compressed (zip) folder**
- On Mac, click **Compress "Lab 3"**

ii. This will create a zip folder on your computer. Rename the zip folder so that it has the following format (replace the below with your first and last name):

```
George-Burdell.zip
```

iii. Upload this `.zip` file to the Lab 3 assignment on **Gradescope.** After that, **you will also need to upload a link to your deployed web app**.

To double check that you did this correctly, please download your submission and run your app to ensure everything works and looks correct.

Please put all images that you used in your website in this Lab 3 folder. Otherwise, your grader will not be able to see images that you included.

2. ***Streamlit Link Submission***

i. Navigate to the top right corner of your deployed Streamlit web app and click on the **Share** button.

ii. Make sure that your app is **PUBLIC** by checking the bottom left corner

iii. Press ***Copy Link*** on the bottom right corner

iv. Paste this link into the appropriate **Gradescope** assignment

Note: Late submissions are **NOT** allowed. Submissions will not be accepted after the due date unless an official excuse is provided by the Dean of Students.

Demos

Visit the Canvas page under the module titled ***Lab Demos*** for more information on demos and demo signups!

Extra Credit

Resume Description or LinkedIn Post Submission (10 points)

Complete **one** of the following assignments for 10 points of extra credit:

- Craft a detailed description of this project, highlighting the key programming skills and specific learning outcomes you have achieved. This description should be tailored for inclusion in your resume, showcasing your hands-on experience and the technical competencies you have developed. For guidance on crafting an effective resume, we recommend consulting the Georgia Tech Career Center's resource guide, available at https://career.gatech.edu/resumes/.
- **OR** share your project experience on LinkedIn. Create a post that not only showcases images from your project but also highlights the skills you have honed and the insights you have gained throughout the lab. This is an excellent opportunity to engage with your network and demonstrate your practical expertise in a real-world application.

Submit either your resume project description or a link to your LinkedIn post on Gradescope for extra credit (one per team).

Point Distribution

Assessment Item	Points
Demo: API Web Page	40
Demo: Chatbot Page	30
Demo: Extra Credit	10
Demo: Questions	30
Total	**110/100**